

Employee Directory Report

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Working UI Screenshot

The application displays the add/edit form, search and sort controls, seeded employee records, active headcount, and active/inactive status badges.

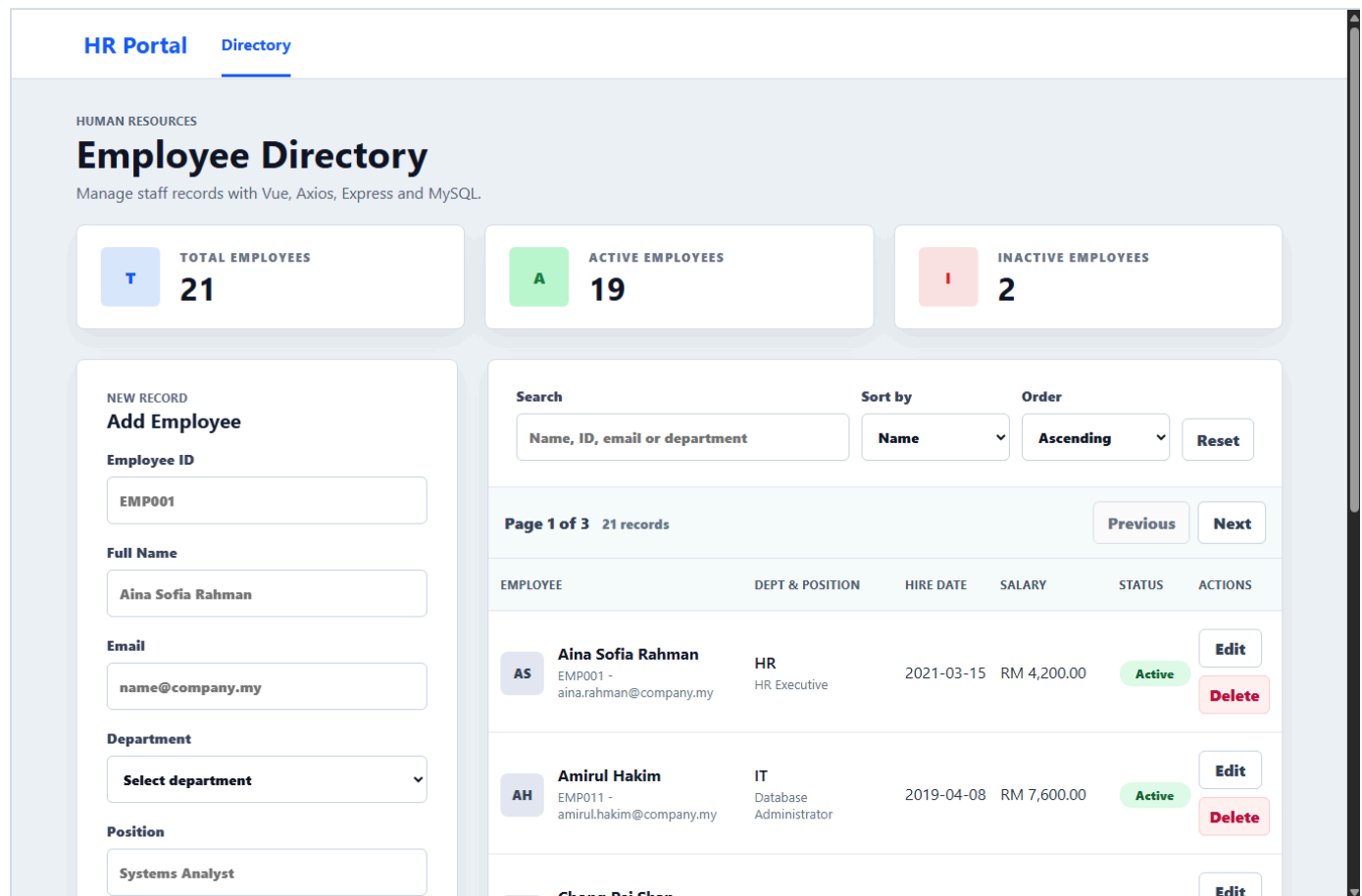


Figure 1: Vue Employee Directory running at <http://127.0.0.1:5174> with MySQL seed data.

Chapter 8 Learning Outcomes

- **LO1: Connect Vue to a backend.** The Vue app calls the Express REST API on port 3001 and keeps the employee list in the parent component state.
- **LO2: Use Axios.** All HTTP requests go through one Axios instance in `src/services/api.js`, including a request logger and a response error mapper.
- **LO3: Build validated Vue forms.** `EmployeeForm.vue` uses `v-model.trim` and `v-model.number`, validates all required fields, and displays inline errors before any request is sent.
- **LO4: Perform CRUD with MySQL.** Express implements create, read, update, delete, server-side search, and whitelisted sorting with prepared statements through the `mysql2` promise pool.

Implementation Notes

The application is separated into form, list, and search/sort components. The SQL script creates the database, table, and 21 sample records across HR, IT, Finance, Marketing, and Operations, with inactive employees included for status testing.

Challenges and Resolutions

- **Safe sorting:** Directly accepting a column name from the client would risk SQL injection. The API maps requested sort keys to a fixed whitelist before building the ORDER BY clause.
- **Consistent errors:** Network and server errors can have different Axios shapes. The response interceptor converts them into one message format used by the UI banner.
- **Responsive layout:** The form and directory use CSS grid and a horizontally safe table layout so the page remains usable on laptop and tablet widths.